

MASS-ENG-008

Document Engine Specification

Page 1 – Purpose & Scope

Purpose

Define the enterprise document subsystem responsible for creating, storing, versioning, rendering, and governing documents throughout their lifecycle.

Objectives

- Centralize document management
- Support templates and generated documents
- Maintain document history and metadata
- Integrate with workflows and knowledge

Out of Scope

Business process orchestration and AI decision-making.

Page 2 – Architecture

Core Components

Document Repository, Template Manager, Rendering Service, Version Manager, Metadata Service, Storage Abstraction Layer.

Relationships

Consumes Enterprise Core, Identity, Security, Event Bus, and Knowledge Engine. Provides document services to all business modules.

Page 3 – Functional Specification

Requirements

1. Create and store documents.
2. Maintain immutable version history.
3. Support reusable templates.
4. Attach metadata and classifications.
5. Publish document lifecycle events.

Non-Functional

Scalable storage, auditability, storage abstraction, high availability, secure access.

Page 4 – Interfaces & Examples

Interfaces

CreateDocument(), UpdateDocument(), RenderDocument(), RetrieveDocument(), ArchiveDocument().

Example Flow

Workflow completes → Document generated from template → Metadata applied → Version stored → Event published → Knowledge index updated.

Page 5 – Construction Package

Claude Checklist

Implement repository; storage abstraction; template engine; rendering service; version manager; metadata integration; event publishing; authorization; unit and integration tests.

Definition of Done

Documents can be securely created, rendered, versioned, retrieved, archived, and integrated with

enterprise workflows.

Constitution References

Knowledge, Security, Governance, Platform Principles, Enterprise Structure.